

Math 184 Notes

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1 Lecture 1

Definition 1.1. Let $f : A \rightarrow B$ be a map.

- We say that f is an injection if $f(a) \neq f(a')$ for $a \neq a' \in A$. We write $f : A \xrightarrow{\text{inj}} B$.
- We say that f is a surjection if, for each $b \in B$, there is $a \in A$ such that $f(a) = b$. We write $f : A \xrightarrow{\text{surj}} B$.
- We say that f is a bijection if it is both an injection and a surjection. We write $f : A \xrightarrow{\text{bij}} B$.

Notation. We write $[n]$ for the set $\{1, 2, \dots, n\}$.

Definition 1.2. A permutation is a bijection $f : [n] \xrightarrow{\text{bij}} [n]$.

Notice that there are $n!$ permutations from $[n]$ to $[n]$.

2 Lecture 2

Suppose that we have n runners, and we want to figure out how many ways there are to award a first, second and third place prize. There are $n(n-1)(n-2)$ ways to do this, as n people can win first, $n-1$ can win second, and $n-2$ can win third. We can generalize this.

Proposition 2.1. *For $k \leq n$, the number of ways to make an ordered list of k elements (without repetitions) of $[n]$ is $(n)_k := \frac{n!}{(n-k)!}$.*

This value $(n)_k$ is simply the number of injections from $[k]$ to $[n]$.

Let us discuss counting principles.

Proposition 2.2 (Addition Principle). *If $A \cap B = \emptyset$, then $|A \sqcup B| = |A| + |B|$.*

Recall that $A \sqcup B$ denotes the disjoint union of A and B .

Proposition 2.3. *If $A \supseteq B$, then $|A \setminus B| = |A| - |B|$.*

For instance, let us find how many positive integers below 1000 have at least two different digits. Let $A = [1000]$ and $B = \{n \in [1000] : n \text{ has the same digits}\}$. Then, $|B| = 9+9+9 = 27$ (for when n has 3, 2 and 1 digits respectively), so that $|A \setminus B| = 1000 - 27 = 973$.

Proposition 2.4 (Product Principle). $|A \times B| = |A| \times |B|$.

For example, the number of k -digit integers is $9 \times 10^{k-1}$, as the first digit must be nonzero, and all other digits can have any digit.

Proposition 2.5. *The number of strings with k letters from an n -letter alphabet is n^k .*

Definition 2.6. For $d \geq 1$, a d -to-1 map is a map $f : A \rightarrow B$ such that, for any $b \in B$, there are exactly d elements of A that map to b .

Note that any d -to-1 map is surjective. However, when $d = 1$, some people mean injective, and others mean bijective.

Proposition 2.7 (Division Principle). *If $f : A \rightarrow B$ is a d -to-1 map, then $|B| = |A|/d$.*

For example, let us say that we have a circular table with n chairs. Let us find the number of ways to seat n people around the circle, where we consider two seatings equivalent if they differ by a rotation. Let A be the set of all permutations of $[n]$, and B be the set of our desirable permutations. Then our relation gives a natural n -to-1 map from A to B , so that $|B| = |A|/n = n!/n = (n-1)!$.

As other example, the number of k elements subsets of $[n]$ is $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. To see this, recall that the number of injections from $[k]$ to $[n]$ is $(n)_k = n!/(n-k)!$. Let A be the set of injections from $[k]$ to $[n]$, and B the set of all k -element subsets of $[n]$. The map from A to B defined by sending f to $\text{im } f$ is a $k!$ -to-1 map, so $|B| = |A|/k! = \binom{n}{k}$.

Theorem 2.8 (Binomial Theorem). *For an integer $n \geq 1$,*

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Proof. Write

$$(x + y)^n = \underbrace{(x + y)(x + y) \cdots (x + y)}_{n \text{ times}}.$$

The number of ways to get $x^k y^{n-k}$ is $\binom{n}{k}$, so this is its coefficient. ■

3 Lecture 3

Proposition 3.1. For all integers $n \geq 1$,

$$2^n = \sum_{k=0}^n \binom{n}{k}.$$

Proof. We have that

$$2^n = (1+1)^n = \sum_{k=0}^n \binom{n}{k} 1^k 1^{n-k} = \sum_{k=0}^n \binom{n}{k}.$$

■

Proposition 3.2. For all integers $n \geq 1$ and $0 \leq k \leq n$,

$$\binom{n}{k} = \binom{n}{n-k}.$$

Proof. We have that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n!}{(n-(n-k))!(n-k)!} = \binom{n}{n-k}.$$

■

Proposition 3.3. For all integers $n \geq 1$, the number of subsets of $[n]$ of odd size is the same as the number of subsets of even size.

Proof. We have that

$$0 = (-1+1)^n = \sum_{k=0}^n \binom{n}{k} (-1)^k 1^{n-k},$$

so that

$$\binom{n}{0} + \binom{n}{2} + \cdots + \binom{n}{2\lfloor n/2 \rfloor} = \binom{n}{1} + \binom{n}{3} + \cdots + \binom{n}{2\lceil n/2 \rceil - 1}.$$

■

Now, let us consider the number of northeast lattice paths from $(0,0)$ to (m,k) (i.e we only move right or up 1 unit each step for $m+k$ steps). We claim this is $\binom{m+k}{k}$. If we label the steps $1, 2, \dots, m+k$ from southwest to northeast, then the numbers of steps going north is a k -element subset of $[m+k]$, so there are $\binom{m+k}{k}$ choices.

Definition 3.4. The n th Catalan number C_n is the number of northeast lattice paths from $(0,0)$ to (n,n) which stay (weakly) above the line $y = x$.

Theorem 3.5. For integers $n \geq 1$,

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

Proof. Let A denote the set of all northeast lattice paths from $(0,0)$ to (n,n) , so that $|A| = \binom{2n}{n}$, and let B be the set of all paths in A which go below the line $x = y$. We wish to find $|B|$. Let C be the set of all northeast lattice paths from $(0,0)$ to $(n+1, n-1)$. We claim that $|B| = |C| = \binom{2n}{n-1}$. Let $P \in B$, and let q be the first lattice point of P that is on the line $y = x - 1$. Reflect the part of P after q over the line $y = x - 1$. This is a northeast path from $(0,0)$ to $(n+1, n-1)$, and this process is clearly reversible, so $|B| = |C|$. Now,

$$C_n = |A \setminus B| = |A| - |B| = \binom{2n}{n} - \binom{2n}{n-1} = \binom{2n}{n} - \frac{n}{n+1} \binom{2n}{n} = \frac{1}{n+1} \binom{2n}{n}.$$

■

4 Lecture 4

Consider the sequences of n '('s and n ')'s, where every '(' must match with a ')'. For example, when $n = 2$, $(())$ is an acceptable sequence, while $))(($ is not. By defining every '(' to be a north step, and every ')' to be a east step, we create a natural bijection with the northeast paths going from $(0, 0)$ to (n, n) that stay weakly above $y = x$. We thus have that the number of such sequences is C_n .

We now discuss more identities.

Proposition 4.1. *If $k < n$, then*

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}.$$

Proof. The right hand side counts all $k + 1$ element subsets of $[n + 1]$. Now, each such set either contains $n + 1$, or does not. If it does contain $n + 1$, then there are $\binom{n}{k}$ such sets, and if it does not contain $n + 1$, then there are $\binom{n}{k+1}$ such sets. Thus, $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$. ■

As an alternative proof, the right hand side counts the paths from $(0, 0)$ to $(n - k, k + 1)$. If the last step is east, then this is the number of paths to $(n - k - 1, k + 1)$, of which there are $\binom{n}{k+1}$, and if the last step is north, this is the number of paths to $(n - k, k)$, of which there are $\binom{n}{k}$.

Proposition 4.2. *For $n \geq 0$,*

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2.$$

Proof. The left hand side counts the lattice paths from $(0, 0)$ to (n, n) . Each of these paths must pass through $(n - k, k)$ for exactly one k . The number of paths passing through this is $\binom{n}{k} \cdot \binom{n}{k} = \binom{n}{k}^2$. Thus, the right hand side also counts the number of lattice paths from $(0, 0)$ to (n, n) . ■

Proposition 4.3.

$$\sum_{k=1}^n k \cdot \binom{n}{k}^2 = n \cdot \binom{2n-1}{n-1}.$$

Proof. The value $k \cdot \binom{n}{k}^2$ counts the paths from $(0, 0)$ to (n, n) passing through $(n - k, k)$, where we require that the path moves up in step ℓ , done over all $1 \leq \ell \leq n$. Thus, the left hand side counts all paths from $(0, 0)$ to (n, n) where we must move up in step ℓ for all possible $1 \leq \ell \leq n$. Now, there are n choices for ℓ . For each of these choices, if we remove the ℓ th step, we obtain a path from $(0, 0)$ to $(n, n - 1)$. Thus, for each ℓ there are $\binom{2n-1}{n-1}$ paths, and as there are n choices for ℓ , the total value is $n \cdot \binom{2n-1}{n-1}$ paths. ■

5 Lecture 5

Notice that $\binom{n}{k}$ is the number of 0–1 strings of length n with k 1's.

Proposition 5.1. *The number of strings with a_i i 's for $1 \leq i \leq k$, where the total length is $a_1 + a_2 + \dots + a_k = n$ is*

$$\binom{n}{a_1, a_2, \dots, a_k} := \frac{n!}{a_1! a_2! \dots a_k!}.$$

Proof. Choose a permutation in $n!$ ways. Replace $1, 2, \dots, a_1$ by $1, a_1 + 1, a_1 + 2, \dots, a_1 + a_2$ by 2 , and so on. This gives us an $a_1! a_2! \dots a_k!$ -to-1 map from the set of permutations of $[n]$ to the set of desired strings. ■

Theorem 5.2 (Multinomial Theorem).

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{a_1 + a_2 + \dots + a_k = n} \binom{n}{a_1, a_2, \dots, a_k} x_1^{a_1} x_2^{a_2} \dots x_k^{a_k}.$$

The proof of this is the same as for the Binomial Theorem.

Now, just as $\binom{a+b}{b}$ is the number of northeast lattice paths from $(0, 0)$ to (a, b) , $\binom{a_1 + a_2 + \dots + a_k}{a_1, a_2, \dots, a_k}$ counts the number of “increasing” lattice paths from $(0, 0, \dots, 0)$ to $(a_1, a_2, \dots, a_k)$ in $\mathbb{Z}^k$.

We now discuss the Pigeonhole Principle. The Pigeonhole Principle states that, if we try to put $m > n$ balls in n boxes, then at least one box will contain at least 2 balls. More generally, we have the following.

Theorem 5.3. *If $A_1, A_2, \dots, A_k$ are pairwise disjoint finite sets, and*

$$|A_1 \sqcup A_2 \sqcup \dots \sqcup A_k| > kr,$$

then $|A_i| > r$ for some $i \in [k]$.

Proof. Suppose that $|A_i| \leq r$ for all $i \in [k]$. Then,

$$|A_1 \sqcup A_2 \sqcup \dots \sqcup A_k| = |A_1| + |A_2| + \dots + |A_k| \leq kr,$$

a contradiction. ■

Proposition 5.4. *Let q be an odd integer not divisible by 5 (i.e. an integer ending in 1, 3, 7, or 9). Then there is a number of the form $11\dots 1$ that is divisible by q .*

Proof. Set $a_i = \underbrace{11\dots 1}_{i \text{ times}}$ for $i \in \mathbb{Z}^+$. Consider the residues of a_i modulo q . Write r_i for the remainder of a_i modulo q . After $q + 1$ steps, we get a_i and a_j , with $i < j$, such that $r_i = r_j$. In other words, $a_j - a_i$ is divisible by q . Now, $a_j - a_i = \underbrace{11\dots 1}_{j-i \text{ times}} \underbrace{00\dots 0}_{i \text{ times}}$. Now, by assumption, q is relatively prime to 10, so that $q \mid \underbrace{11\dots 1}_{j-i \text{ times}} \underbrace{00\dots 0}_{i-1 \text{ times}}$. Repeating this, $q \mid \underbrace{11\dots 1}_{j-i \text{ times}}$. ■

Proposition 5.5. *Choose $n + 1$ distinct elements of $[2n]$. Then,*

(1) Two of them sum up to $2n + 1$.

(2) Two of them differ by n .

Proof. Let $A_i = \{i, 2n + 1 - i\}$ for $i = 1, 2, \dots, n$. Out of the $n + 1$ chosen integers, two must belong to some A_i . These two integers sum to $2n + 1$, so we are done. Now, let $B_i = \{i, n + i\}$. Once again, two of our $n + 1$ integers must belong to the same B_j , so their difference is n . ■

Proposition 5.6. *For each $p \in \mathbb{R}$ and $\varepsilon > 0$, there are integers a and b such that $|p - a/b| < \varepsilon$.*

Proof. Choose $N \in \mathbb{Z}$ such that $1/N < \varepsilon$, and replace ε with $1/N$. Let $B_i = [\frac{i-1}{N}, \frac{i}{N})$ for $i \in [N]$. Then, $\bigsqcup_{i=1}^N B_i = [0, 1)$. For each $n \geq 1$, let $a_n \in [0, 1)$ be given by $a_n = np - \lfloor np \rfloor$. By the pigeonhole principle, there are some a_n and a_m belonging to the same B_i . Then, $|a_n - a_m| < \varepsilon$. Thus, $|(np - mp) - (\lfloor np \rfloor - \lfloor mp \rfloor)| < \varepsilon$, so that

$$\left| p - \frac{\lfloor np \rfloor - \lfloor mp \rfloor}{n - m} \right| < \frac{\varepsilon}{|n - m|} < \varepsilon.$$

■

6 Lecture 6

Definition 6.1. A weak composition of n is a sequence $(a_1, \dots, a_k)$ such that each $a_i \geq 0$ and $a_1 + a_2 + \dots + a_k = n$.

Proposition 6.2. The number of weak compositions of n is $\binom{n+k-1}{n}$.

Proof. By symmetry, $\binom{n+k-1}{n} = \binom{n+k-1}{k-1}$. This counts the subsets of $[n+k-1]$ of size $k-1$. Suppose that we have $n+k-1$ empty spots. If we place n balls into these spots, leaving $k-1$ empty spots, and place a wall in each of the $k-1$ empty spots. Now, there are k spaces between the walls, so if $(a_1, \dots, a_k)$ records the number of balls between each pair of walls, we have a bijection between $k-1$ element subsets of $[n+k-1]$ and weak compositions of n . ■

Definition 6.3. A multiset over S is a collection of elements of S , where repetitions are allowed, and order does not matter. The size of a multiset is the number of elements in it counted with multiplicities.

We claim that multisets over $[k]$ of size n are in bijection with weak compositions of n into k parts. Given $(a_1, a_2, \dots, a_k)$, we create a multiset of $[k]$ containing $i \in [k]$ a_i times. As a corollary of this, the number of multisets over $[k]$ of size n is $\binom{n+k-1}{n}$.

Definition 6.4. A weak composition $(a_1, a_2, \dots, a_k)$ is a composition if each $a_i > 0$.

Proposition 6.5. The number of compositions of n into k parts is $\binom{n-1}{k-1}$.

Proof. We have a bijection between compositions of n into k parts and weak compositions of $n-k$ into k parts by $(a_1, a_2, \dots, a_k) \mapsto (a_1-1, a_2-1, \dots, a_k-1)$. Thus, the number of compositions of n into k parts is $\binom{n-k+k-1}{k-1} = \binom{n-1}{k-1}$. ■

Notice that the number of compositions of n into an arbitrary number of parts is $\sum_{k=1}^n \binom{n-1}{k-1} = 2^{n-1}$.

Definition 6.6. $A = \{A_1, A_2, \dots, A_k\}$ is a partition of $[n]$ into k blocks if $A_1, A_2, \dots, A_k$ are disjoint subsets of $[n]$ with $A_1 \sqcup A_2 \sqcup \dots \sqcup A_k = [n]$, where the order of the A_i and the order of the elements of the A_i do not matter.

Definition 6.7. The Stirling number of the second kind is $S(n, k)$, which denotes the number of partitions of $[n]$ into k parts.

For example, for $n > 0$, $S(n, 0) = 0$, $S(0, 0) = 1$, and $S(n, k) = 0$ for $n < k$.

Proposition 6.8. For each $n \geq k > 0$,

$$S(n, k) = S(n-1, k-1) + k \cdot S(n-1, k).$$

Proof. We have that $S(n, k)$ is the partitions of $[n]$ into k blocks. Look at the block A_i containing n . Either $A_i = \{n\}$ or $A_i = \{n\} \sqcup B_i$ for $B_i \neq \emptyset$. In the first case, we can remove n to get a partition of $[n-1]$ into $k-1$ blocks. In the second case, we can remove n from A_i to get a partition of $[n-1]$ into k blocks. However, this is a k -to-1 map, as if we are given $A_1, \dots, A_k$ partitioning $[n-1]$, we can add n to any of the A_i 's. Thus, $S(n, k) = S(n-1, k-1) + k \cdot S(n-1, k)$. ■

7 Lecture 7

Proposition 7.1.

$$S(n, k) = \sum_{1 \leq a_1 \leq a_2 \leq \dots \leq a_{n-k} \leq k} a_1 \cdot a_2 \cdot \dots \cdot a_{n-k}.$$

Proof. Denote the right hand side by $H(n, k)$. We want to show that $H(n, k) = H(n-1, k-1) + k \cdot H(n-1, k)$. Suppose that $a_{n-k} < k$. This contributes $H(n-1, k-1)$, as we can replace k by $k-1$ and preserve the length of the sequence as $n-k$. If $a_{n-k} = k$, we get a shorter sequence by removing it, and the sum of these sequences is $H(n-1, k)$. Since $a_{n-k} = k$, we multiply it by k . Now, if $k = 1$, then all terms are 1, so $H(n, 1) = 1$, and when $n = k$, there are no such sequences, so the sum is 1, so that $H(n, k)$ has the same initial conditions as $S(n, k)$. ■

Recall that we write $(x)_k = x \cdot (x-1) \cdot \dots \cdot (x-k+1)$, so that if $x \in \mathbb{Z}_{>0}$, $(x)_k$ is the number of injections from $[k]$ to $[x]$.

Theorem 7.2. For all $x \in \mathbb{R}$ and $n \in \mathbb{Z}_{\geq 1}$,

$$x^n = \sum_{k=0}^n S(n, k)(x)_k.$$

Proof. Suppose that x is a positive integer. Then x^n is the number of words of length n in the alphabet $[x]$. Given a word, place the locations of i in A_i , so that $A_1, \dots, A_x$ partition $[n]$. Let k be the number of distinct letters in our word. Then we have a partition of $[n]$ into k blocks. This is an $(x)_k$ -to-1 map. Thus, for each k , the number of words using exactly k different letters is $S(n, k)(x)_k$. Summing over k gives the result for positive integers x . Now, we have shown that x^n and $\sum_{k=0}^n S(n, k)(x)_k$ are two polynomials agreeing at infinitely many values of x , and so they must be the same polynomial. ■

Proposition 7.3.

$$S(n+1, k) = \sum_{i=0}^n \binom{n}{i} S(n-i, k-1).$$

Proof. $S(n+1, k)$ counts partitions of $[n+1]$ into k blocks. Let $i+1$ be the size of the block containing $n+1$. We can choose the remaining i elements of this in $\binom{n}{i}$ ways. The remaining blocks have $S(n-i, k-1)$ choices. ■

Definition 7.4. The Bell Number is

$$B(n) = \sum_{k=0}^n S(n, k).$$

Notice that $B(n)$ counts all possible partitions of $[n]$.

Proposition 7.5.

$$B(n+1) = \sum_{k=0}^n \binom{n}{k} B(k).$$

Proof. Choose the block of size $i + 1$ containing $n + 1$ in $\binom{n}{i} = \binom{n}{n-i}$ ways. The remaining blocks can be chosen in $B(n - i)$ ways. Thus,

$$B(n + 1) = \sum_{i=1}^n \binom{n}{i} B(n - i) = \sum_{i=1}^n \binom{n}{n-i} B(n - i) = \sum_{i=1}^n \binom{n}{i} B(i).$$

■

8 Lecture 8

The number $S(n, k)$ is the number of partitions of n into k blocks. Each of these can be turned into a surjection from $[n]$ to $[k]$ by sending $x \in A_i$ to i . However, this is a $k!$ -to-1 map, so we have that

$$S(n, k) = \frac{\text{The number of surjections from } [n] \text{ to } [k]}{k!}.$$

Now, if we can find a closed form for the numerator, we will be able to find a closed form for $S(n, k)$. We use the Inclusion-Exclusion principle.

Proposition 8.1 (Inclusion-Exclusion Principle). *For any finite sets $A_1, \dots, A_n$,*

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{j=1}^n (-1)^{j-1} \sum_{\substack{A \subseteq [n] \\ |A|=j}} \left| \bigcap_{\ell \in A} A_\ell \right|.$$

Proof. Suppose that $x \in A_1 \cup A_2 \cup \dots \cup A_n$, and suppose that it belongs to exactly k of the A_i 's. For each j , x belongs to exactly $\binom{k}{j}$ terms in the right hand side. In other words, the contribution of x to the right hand side is

$$\sum_{j=1}^k (-1)^{j-1} \binom{k}{j},$$

which is 1 by the binomial theorem. ■

Theorem 8.2. *The number of surjections $f : [n] \rightarrow [k]$ is given by*

$$\sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n.$$

Proof. Let $A_1, \dots, A_k$ be given by

$$A_i = \{f : [n] \rightarrow [k] : f^{-1}(i) = \emptyset\}.$$

By the subtraction principle,

$$S(n, k) \cdot k! = k^n - \left| \bigcup_{i=1}^k A_i \right|.$$

Now,

$$\left| \bigcup_{i=1}^k A_i \right| = \sum_{j=1}^k (-1)^{j-1} \sum_{\substack{A \subseteq [k] \\ |A|=j}} \left| \bigcap_{\ell \in A} A_\ell \right|.$$

If $|A| = j$, then

$$\left| \bigcup_{i \in A} A_i \right| = \{f : [n] \rightarrow [k] : f^{-1}(A) = \emptyset\} = (k-j)^n.$$

Thus,

$$S(n, k) \cdot k! = (-1)^0 \binom{k}{0} (k-0)^n - \sum_{j=1}^k (-1)^{j-1} \sum_{\substack{A \subseteq [n] \\ |A|=j}} (k-j)^n = \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n.$$

■

As a corollary,

$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n.$$

Definition 8.3. A permutation $f : [n] \rightarrow [n]$ is a derangement if $f(i) \neq i$ for all $i \in [n]$. We write $D(n)$ for the number of derangements.

Proposition 8.4.

$$D(n) = n! \cdot \sum_{j=0}^n \frac{(-1)^j}{j!}.$$

Proof. Let $A_1, \dots, A_n$ be given by $A_i = \{f : [n] \rightarrow [n] : f \text{ bijective}, f(i) = i\}$. Now, for $|A| = j$, we have that

$$\left| \bigcap_{i \in A} A_i \right| = (n-j)!.$$

Thus,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{j=1}^n (-1)^{j-1} \binom{n}{j} (n-j)! = \sum_{j=1}^n (-1)^{j-1} \frac{n!}{j!}.$$

Thus,

$$D(n) = n! - \left| \bigcup_{i=1}^n A_i \right| = n! \sum_{j=0}^n \frac{(-1)^j}{j!}.$$

■

A corollary of this is that

$$\lim_{n \rightarrow \infty} \frac{D(n)}{n!} = \sum_{j=0}^{\infty} (-1)^j \frac{1}{j!} = \frac{1}{e}.$$

9 Lecture 9

Suppose that we want to find the best candidate out of n candidates, where we have to accept or reject each candidate immediately after interviewing them, we cannot call them back. The optimal strategy is to reject the first n/e candidates, and then hire the first person who is better than everyone before them. The probability of success is about $1/e$.

Definition 9.1. A partition of n is a weakly decreasing sequence $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \lambda_k > 0)$ of positive integer such that $n = |\lambda| := \lambda_1 + \lambda_2 + \cdots + \lambda_k$. We write $\lambda \vdash n$.

Write $p(n)$ for the number of partitions of n (i.e. $|\{\lambda \vdash n\}|$). We write $\ell(\lambda)$ for the number of nonzero parts of λ , and $p(n, k)$ for the number of partitions of n of length k (i.e. $|\{\lambda \vdash n : \ell(\lambda) = k\}|$).

Proposition 9.2. $p(n, k) = p(n - 1, k - 1) + p(n - k, k)$.

Proof. If $\lambda_k = 1$, then remove it to obtain a partition $n - 1$ into $k - 1$ parts. There are $p(n - 1, k - 1)$ ways to do this. Otherwise, all parts are at least 2, so we can decrease all parts by 1 to get a partition of $n - k$ into k parts, which there are $p(n - k, k)$ of. ■

We will usually set $\lambda_i = 0$ for $i > \ell(\lambda)$, so that we consider $(4, 2, 1, 1)$ to be the same as $(4, 2, 1, 1, 0, 0, 0)$. For each partition, place n boxes so that there are λ_i boxes in row i , aligned to the left (this is called a Young Diagram). Now, for a partition λ , we can get a conjugate partition λ' by reflecting the corresponding Young Diagram across the main diagonal (in the sense of matrices). For example, if $\lambda = (5, 3, 1, 1)$, then we get $\lambda' = (4, 2, 2, 1, 1)$ (draw it out).

Proposition 9.3. For each $k \leq n$,

$$|\{\lambda \vdash n : \ell(\lambda) \geq k\}| = |\{\lambda \vdash n : \lambda_1 \geq k\}|.$$

Proposition 9.4. For all n ,

$$|\{\lambda \vdash n : \lambda_1 = \lambda_2\}| = |\{\lambda \vdash n : \lambda_i \geq 2\}|.$$

Both of these follow easily by conjugations.

Let $p_{d,even}(n)$ denote the number of partitions of n into an even number of distinct parts. Similarly, let $p_{d,odd}(n)$ denote the number of partitions of n into an odd number of distinct parts.

Theorem 9.5 (Euler Pentagonal Theorem). For all n ,

$$p_{d,even}(n) - p_{d,odd}(n) = \begin{cases} (-1)^j & n = \frac{j(3j \pm 1)}{2} \\ 0 & \text{else} \end{cases}.$$

Proof. Let $\lambda \vdash n$ into distinct parts, and draw the corresponding Young Diagram. Let L_λ be the bottom row of this diagram, and let D_λ be the last boxes in each row i such that $\lambda_i = \lambda_i - i + 1$. Define $\varphi(\lambda)$ to be given by another Young Diagram constructed as follows: if $|D_\lambda| > |L_\lambda|$, then move L_λ to the right of D_λ , and then move them to the left. If $|D_\lambda| \leq |L_\lambda|$,

move D_λ into a new bottom row. Notice that $\varphi(\varphi(\lambda)) = \lambda$. Now, φ changes the parity of $\ell(\lambda)$. However, if $\lambda = (2j-1, 2j-2, \dots, j)$ or $(2j, 2j-1, \dots, j+1)$, then $\varphi(\lambda)$ is not a valid partition into distinct parts (drawing it out makes these clear). These give that n is $\frac{j(3j-1)}{2}$ and $\frac{j(3j+1)}{2}$ respectively. Since the extra Young Diagram has height j , the difference in this case is $(-1)^j$. ■

10 Lecture 10

Proposition 10.1. *The number of partitions of n into odd parts is the number of partitions of n into distinct parts.*

Proof. Let λ be a partition of n into odd parts. For each $j \geq 0$, suppose that $2j + 1$ appears r_j times in λ . Define a partition μ into distinct parts as follows: μ contains $(2j + 1)2^k$ if and only if 2^k appears in the binary expansion of r_j . By construction, μ partitions n , and this is a bijection. ■

Suppose that we have n distinct balls and k distinct boxes. We discuss the following three questions.

- (1) How many ways are there to put the balls into the boxes?
- (2) How many ways are there to put the balls into the boxes so that no two balls occupy the same box?
- (3) How many ways are there to put the balls into the boxes so that each box contains at least one ball?

Now, the first question is the number of maps from $[n]$ to $[k]$, the second question is the number of injections from $[n]$ to $[k]$, and the last question is the number of surjections from $[n]$ to $[k]$.

We can also consider balls to be indistinguishable in the sense that we consider maps $f : [n] \rightarrow [k]$ to be the same under the relation $f \sim g$ if there is a permutation $h : [n] \rightarrow [n]$ such that $f = g \circ h$. Similarly, we can consider boxes to be indistinguishable by considering maps $f : [n] \rightarrow [k]$ to be the same under $f \sim g$ if there is a permutation $h : [k] \rightarrow [k]$ such that $f = h \circ g$.

We aim to fill out the following table, where the blank boxes indicate the number of functions of the type in the column.

Balls $[n]$	Boxes $[k]$	Any f	Injective f	Surjective f
dist	dist			
indist	dist			
dist	indist			
indist	indist			

We already know this partial solution.

Balls $[n]$	Boxes $[k]$	Any f	Injective f	Surjective f
dist	dist	k^n	$(k)_n$	$k! \cdot S(n, k)$
indist	dist			
dist	indist			
indist	indist			

Now, if the balls are indistinguishable and the boxes are not, the total number of functions is the number of multisets over $[k]$ of size n , which is $\binom{n+k-1}{n}$. The total number of injective functions is $\binom{k}{n}$. The total number of surjective functions is the number of multisets over $[k]$ of size $n - k$, which is $\binom{n-1}{n-k}$. This gives us the following.

Balls $[n]$	Boxes $[k]$	Any f	Injective f	Surjective f
dist	dist	k^n	$(k)_n$	$k! \cdot S(n, k)$
indist	dist	$\binom{n+k-1}{n}$	$\binom{k}{n}$	$\binom{n-1}{n-k}$
dist	indist			
indist	indist			

Suppose that the boxes are distinguishable and the balls are not. The total number of functions is the number of partitions of $[n]$ into at most k parts. The total number of injective functions is exactly 1 (assuming $n \leq k$), as which boxes contain balls does not matter. If $n > k$, this is 0. The number of surjective maps is the number of partitions of $[n]$ into k parts, which is $S(n, k)$.

Balls $[n]$	Boxes $[k]$	Any f	Injective f	Surjective f
dist	dist	k^n	$(k)_n$	$k! \cdot S(n, k)$
indist	dist	$\binom{n+k-1}{n}$	$\binom{k}{n}$	$\binom{n-1}{n-k}$
dist	indist	$\sum_{i=0}^k S(n, i)$	$\begin{cases} 1 & n \leq k \\ 0 & \text{else} \end{cases}$	$S(n, k)$
indist	indist			

Now suppose that both the balls and boxes are distinct. The number of functions is the number of partitions of n into at most k parts (this is different than partitions of $[n]$). The number of injective functions is the same as above for the same reason. The number of surjective functions is the number of partitions of n into k parts.

Balls $[n]$	Boxes $[k]$	Any f	Injective f	Surjective f
dist	dist	k^n	$(k)_n$	$k! \cdot S(n, k)$
indist	dist	$\binom{n+k-1}{n}$	$\binom{k}{n}$	$\binom{n-1}{n-k}$
dist	indist	$\sum_{i=0}^k S(n, i)$	$\begin{cases} 1 & n \leq k \\ 0 & \text{else} \end{cases}$	$S(n, k)$
indist	indist	$\sum_{i=0}^k p(n, k)$	$\begin{cases} 1 & n \leq k \\ 0 & \text{else} \end{cases}$	$p(n, k)$

11 Lecture 11

Definition 11.1. A formal power series is a sum of the form

$$F(x) = \sum_{n=0}^{\infty} f_n x^n,$$

where $f_0, f_1, \dots$ are a sequence of complex numbers.

We can also add power series as

$$\left(\sum_{n=0}^{\infty} a_n x^n \right) + \left(\sum_{n=0}^{\infty} b_n x^n \right) = \sum_{n=0}^{\infty} (a_n + b_n) x^n$$

and multiply them as

$$\left(\sum_{n=0}^{\infty} a_n x^n \right) \left(\sum_{n=0}^{\infty} b_n x^n \right) = \sum_{n=0}^{\infty} \left(\sum_{i=0}^n a_i b_{n-i} \right) x^n.$$

These formal power series form a ring, denoted $\mathbb{C}[[x]]$.

Definition 11.2. If $F(x)$ and $G(x)$ are two power series such that $F(x) \cdot G(x) = 1$, then we say $G(x) = F(x)^{-1} = \frac{1}{F(x)}$.

Let us show that $F(x) = 1 - x$ has an inverse in $\mathbb{C}[[x]]$. Let $G(x) = 1 + x + x^2 + \dots$. We have that

$$F(x)G(x) = (1 - x)(1 + x + x^2 + \dots) = 1 + (1 - 1)x + (1 - 1)x^2 + \dots = 1.$$

However, $F(x) = x$ does not have an inverse in $\mathbb{C}[[x]]$.

Definition 11.3. Given a sequence $f_0, f_1, f_2, \dots$, we define the ordinary generating function (ogf) as

$$\sum_{n=0}^{\infty} f_n x^n$$

and the exponential generating function (egf) as

$$\sum_{n=0}^{\infty} \frac{f_n}{n!} x^n.$$

Notice that

$$\left(\sum_{n=0}^{\infty} \frac{a_n}{n!} x^n \right) \left(\sum_{n=0}^{\infty} \frac{b_n}{n!} x^n \right) = \sum_{n=0}^{\infty} \left(\sum_{i=0}^n \frac{a_i}{i!} \frac{b_{n-i}}{(n-i)!} \right) x^n = \sum_{n=0}^{\infty} \frac{\sum_{i=0}^n \binom{n}{i} a_i b_{n-i}}{n!}$$

Let us investigate if

$$F(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

as an inverse. Consider

$$G(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}.$$

We have that

$$F(x)G(x) = \sum_{n=0}^{\infty} \left(\sum_{i=0}^n \binom{n}{i} (-1)^{n-i} \right) \frac{x^n}{n!} = 1.$$

12 Lecture 12

Suppose that $f(x) = \sum_{n=0}^{\infty} f_n x^n$. We write $[x^n]f(x) = f_n$ to be the coefficient of x^n . If $F(x) = \sum_{n=0}^{\infty} f_n x^n / n!$, we write $[x^n/n!]F(x) = f_n$.

Given a formal power series

$$F(x) = \sum_{n=0}^{\infty} f_n x^n,$$

we define

$$F'(x) = \sum_{n=1}^{\infty} n \cdot f_n x^{n-1}.$$

This satisfies the usual rules of differentiation.

For $\alpha \in \mathbb{R}$, we have that

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n,$$

where

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}.$$

This satisfies all the usual properties.

Consider

$$F(x) = \sum_{n=0}^{\infty} (n+1)x^n = 1 + 2x + 3x^2 + \cdots.$$

We have that

$$F(x) - xF(x) = \sum_{n=0}^{\infty} (n+1-n)x^n = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x},$$

so that $F(x) = 1/(1-x)^2$.

Suppose that $a_0 = a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$, and let

$$F(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Let us try to find a closed formula for $f(x)$. We have that

$$F(x) - xF(x) - x^2F(x) = a_0 = 1,$$

so that

$$F(x) = \frac{1}{1-x-x^2}.$$

We can write

$$F(x) = \frac{1}{\sqrt{5}} \left(\frac{1}{x+\tau} - \frac{1}{x+\bar{\tau}} \right),$$

where

$$\tau = \frac{1+\sqrt{5}}{2} \quad \bar{\tau} = -\frac{1}{\tau} = \frac{1-\sqrt{5}}{2}.$$

Now, we have that

$$\frac{1}{x + \tau} = \frac{1}{\tau} \frac{1 + x/\tau}{1} = \frac{1}{\tau} (1 + x\bar{\tau} + x^2\bar{\tau}^2 + \dots),$$

and

$$\frac{1}{x + \bar{\tau}} = \frac{1}{\bar{\tau}} \frac{1}{1 + x/\bar{\tau}} = \frac{1}{\bar{\tau}} (1 + x\tau + x^2\tau^2 + \dots).$$

We thus have that

$$\begin{aligned} a_n &= \frac{1}{\sqrt{5}} \left(\frac{\bar{\tau}^n}{\tau} - \frac{\tau^n}{\bar{\tau}} \right) = \frac{1}{\sqrt{5}} \left(\frac{\tau^{n+1} - \bar{\tau}^{n+1}}{\tau\bar{\tau}} \right) = \frac{1}{\sqrt{5}} (\bar{\tau}^{n+1} - \tau^{n+1}) \\ &= \frac{1}{\sqrt{5}} \left(\left(\frac{1 - \sqrt{5}}{2} \right)^{n+1} - \left(\frac{1 + \sqrt{5}}{2} \right)^{n+1} \right). \end{aligned}$$

Now, let $F(x) = \sum_{n=0}^{\infty} C_n x^n$. Firstly, we have that

$$C_{n+1} = \sum_{i=0}^n C_i \cdot C_{n-i}.$$

To see this, let $q = (i+1, i+1)$ be the first point on $y = x$ that a path touches. Then, as the path before touching q must be east, and the first step must be north, so there are $C_i \cdot C_{n-i}$ paths passing through q . Now, we have that

$$x^{n+1} C_{n+1} = \sum_{i=0}^n C_i C_{n-i} x^{n+1} = x \sum_{i=0}^n (C_i x^i) (C_{n-i} x^{n-i}).$$

Summing over all n , we have that

$$F(x) - 1 = x \cdot F(x)^2$$

“Solving”, we have that

$$F(x) = \frac{1 \pm \sqrt{1 - 4x}}{2x},$$

though the right hand side does not really make sense as a formal power series.

13 Lecture 13

Using the binomial theorem, we have that

$$\sqrt{1-4x} = \sum_{n=0}^{\infty} \binom{1/2}{n} (-4)^n x^n,$$

which is a valid formal power series. However, $2x$ is not invertible, so we must say

$$2x \cdot F(x) - 1 = \pm \sqrt{1-4x},$$

where both sides make sense as formal power series. Comparing coefficients, we see that we must take the negative branch. Now, if we square both sides, we get that

$$4x^2 F^2(x) - 4xF(x) + 1 = 1 - 4x,$$

so that we get

$$xF^2(x) - F(x) = -1,$$

which has been established. Working backwards, we get our original solution, choosing the negative branch (rigorously).

Definition 13.1. Let $F(x) = \sum_{n=0}^{\infty} a_n x^n$. We define the minimal degree of F to be the least n with $a_n \neq 0$. We write $\text{mindeg } F(x)$.

Definition 13.2. Let $F(x) = \sum_{n=0}^{\infty} a_n x^n$, and let $F_1(x), F_2(x), \dots$ be a sequence of formal power series. We say that this sequence stabilizes to $F(x)$ if, for each $n \geq 0$, the sequence $[x^n]F_1(x), [x^n]F_2(x), \dots$ stabilizes to a_n . We write $\lim_{j \rightarrow \infty} F_j(x) = F(x)$.

Definition 13.3. We write $\sum_{j=0}^{\infty} F_j(x) = F(x)$ if

$$\lim_{m \rightarrow \infty} \left[\sum_{j=0}^m F_j(x) \right] = F(x).$$

Definition 13.4. If $G_j(0) = 0$ for $j \geq 0$, then we say

$$\prod_{j=0}^{\infty} (1 + G_j(x)) = F(x)$$

if

$$\lim_{m \rightarrow \infty} \left[\prod_{j=0}^m (1 + G_j(x)) \right] = F(x).$$

Proposition 13.5. $\sum_{j=0}^{\infty} F_j(x)$ stabilizes if and only if $\lim_{j \rightarrow \infty} \text{mindeg } F_j(x) = \infty$. If $G_j(0) = 0$ for $j \geq 0$, then $\prod_{j=0}^{\infty} (1 + G_j(x))$ stabilizes if and only if $\lim_{j \rightarrow \infty} \text{mindeg } G_j(x) = \infty$.

If $F(0) = 0$ and $\alpha \in \mathbb{C}$, we define

$$(1 + F(x))^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} F(x)^n.$$

As $\text{mindeg } F(x)^n \geq n$, this stabilizes.

Let us try to find a formula for

$$F(x) = \sum_{n=0}^{\infty} p(n)x^n.$$

Proposition 13.6.

$$F(x) = \prod_{j=1}^{\infty} \frac{1}{1 - x^j}.$$

Proof. We have that

$$\frac{1}{1 - x^j} = 1 + x^j + x^{2j} + x^{3j} + \dots,$$

so this becomes

$$(1 + x + x^2 + \dots)(1 + x^2 + x^4 + \dots)(1 + x^3 + x^6 + \dots).$$

We need to check that this stabilizes. Write $\frac{1}{1-x^j} = 1 + G$. Then $\text{mindeg } G = j$, so this stabilizes. Now, for each $\lambda \vdash n$, if m_j is the multiplicity of j , we have that $x^n = x^{m_1 \times 1} x^{m_2 \times 2} x^{m_3 \times 3} \dots$, so we can pick $m_j \times j$ out of the expansion of $1/(1 - x^j)$. This is clearly a bijection, so we have that the coefficient of x^n is $p(n)$. ■

14 Lecture 14

We can generalize the idea of the last lecture to get that

$$\sum_{n=0}^{\infty} p_d(n)x^n = \prod_{j=1}^{\infty} (1 + x^j)$$

and that

$$\sum_{n=0}^{\infty} p_{\text{odd}}(n)x^n = \prod_{j=1}^{\infty} \frac{1}{1 - x^{2j-1}}.$$

Now, we have that $p_{\text{odd}}(n) = p_d(n)$, as seen before. This can also be justified, as

$$1 + x^j = \frac{1 - x^{2j}}{1 - x^j},$$

so

$$\prod_{j=1}^{\infty} (1 + x^j) = \frac{1 - x^2}{1 - x} \cdot \frac{1 - x^4}{1 - x^2} \cdot \frac{1 - x^6}{1 - x^3} \cdots = \frac{1}{1 - x} \cdot \frac{1}{1 - x^3} \cdot \frac{1}{1 - x^5} \cdots = \prod_{j=1}^{\infty} \frac{1}{1 - x^{2j-1}}.$$

Suppose that $f(n)$ counts the number of ways to perform a task on $[n]$ and that $g(n)$ counts the number of ways to perform a second task on $[n]$. Let $h(n)$ denote the number of ways to write $[n] = S \sqcup T$, followed by performing the first task on S and the second task on T . Then,

$$h(n) = \sum_{i=0}^n \binom{n}{i} f(i)g(n-i),$$

so the exponential generating functions satisfy

$$H(x) = F(x) \cdot G(x).$$

Proposition 14.1. *Suppose that we have a sequence of functions $f_1(n), f_2(n), \dots, f_k(n)$, and that*

$$h(n) = \sum_{T_1 \sqcup T_2 \sqcup \cdots \sqcup T_k = [n]} f_1(|T_1|) \cdot f_2(|T_2|) \cdots f_k(|T_k|).$$

Then the exponential generating functions satisfy

$$H(x) = F_1(x) \cdot F_2(x) \cdots F_k(x).$$

Let us find how many ways there are to paint elements of $[n]$ into colors a, b, c, d so that there are an odd number of a 's and an even number of b 's. Let $h(n)$ denote the answer. Then,

$$h(n) = \sum_{T_a \sqcup T_b \sqcup T_c \sqcup T_d = [n]} f_a(|T_a|) \cdot f_b(|T_b|) \cdot f_c(|T_c|) \cdot f_d(|T_d|),$$

where

$$f_a(n) = \begin{cases} 1 & n \equiv 1 \pmod{2} \\ 0 & n \equiv 0 \pmod{2} \end{cases} \quad f_b(n) = \begin{cases} 0 & n \equiv 1 \pmod{2} \\ 1 & n \equiv 0 \pmod{2} \end{cases} \quad f_c(n) = f_d(n) = 1.$$

We have that

$$F_a(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} = \frac{e^x - e^{-x}}{2}$$

and

$$F_b(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = \frac{e^x + e^{-x}}{2}.$$

We have that $F_c(x) = F_d(x) = e^x$, so

$$H(x) = \frac{e^x - e^{-x}}{2} \cdot \frac{e^x + e^{-x}}{2} \cdot e^x \cdot e^x = \frac{1}{4}(e^{4x} - 1) = \sum_{n=1}^{\infty} 4^{n-1} \frac{x^n}{n!}.$$

Thus,

$$h(n) = 4^{n-1}.$$

15 Lecture 15

Suppose that $F(x)$ and $G(x)$ are formal power series with $f(0) = 0$. Then the composition $G(F(x)) = \sum_{n=0}^{\infty} f(n)F(x)^n$ is a well-defined formal power series. This is because the infinite sum stabilizes, as F does not have a constant term.

Let $\Pi([n])$ be the set of partitions of $[n]$, and recall that $B(n) = |\Pi(n)|$.

Theorem 15.1 (Exponential Formula). *Let $f(n)$ be a sequence with $f(0) = 0$. Define $h(0) = 1$ and $h(n) = \sum_{\{S_1, \dots, S_k\} \in \Pi([n])} f(|S_1|) \cdots f(|S_k|)$. Then the exponential generating functions satisfy*

$$H(x) = \sum_{n=0}^{\infty} \frac{F(x)^n}{n!}.$$

Proof. Suppose that $k \geq 0$ is fixed. Then the exponential generating function would be given by $F(x)^k/k!$: $F(x)^k$ is the exponential generating function of $\sum_{T_1 \sqcup \dots \sqcup T_k = [n]} f(|T_1|) \cdots f(|T_k|)$ counting ordering, so we must divide by $k!$ (notice that, although $F(x)^k$ allows the T_i to be empty, we have that $f(0) = 0$, so we are okay). ■

For example, if $f(n) = \begin{cases} 0 & n = 0 \\ 1 & \text{else} \end{cases}$, then we have that

$$h(n) = B(n),$$

and that

$$F(x) = \sum_{n=0}^{\infty} f(n) \frac{x^n}{n!} = e^x - 1.$$

Thus,

$$H(x) = \sum_{n=0}^{\infty} \frac{(e^x - 1)^n}{n!} = e^{e^x - 1}.$$

Let us find the number $h(n)$ of ways to split n people into nonempty groups and have each group sit around a circular table. To do this, we find $H(x) = \sum_{n=0}^{\infty} h(n)x^n/n!$. If we define $f(n) = (n-1)!$ (and $f(0) = 0$) to be the number of ways to set n people in a circular table, we have that

$$F(x) = \sum_{n=0}^{\infty} f(n) \frac{x^n}{n!} = \sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x).$$

Now,

$$H(x) = \exp(F(x)) = e^{-\ln(1-x)} = \frac{1}{1-x} = 1 + x + x^2 + \cdots = \sum_{n=0}^{\infty} n! \cdot \frac{x^n}{n!}.$$

Thus, $h(n) = n!$.

16 Lecture 16

Theorem 16.1 (Compositional Formula). *Let $f(n)$ is a sequence with $f(0) = 0$ and $g(n)$ be a sequence with $g(0) = 1$. If $h(n)$ satisfies $h(0) = 1$ and*

$$h(n) = \sum_{\{S_1, \dots, S_k\} \in \Pi([n])} f(|S_1|) \cdots f(|S_k|) \cdot g(k)$$

for $n \geq 1$, then the exponential generating functions satisfy

$$H(x) = G(F(x)).$$

Proof. This is the same as the proof of the exponential formula. ■

Now, let us find $h(n)$ the number of ways to split n people into nonempty groups, have each group sit around a circular table, and choose to serve a red or white wine at each table. Let $f(0) = 0$ and $f(n) = (n-1)!$, and $g(n) = 2^n$. We have that

$$F(x) = -\ln(1-x) \quad g(x) = \exp(2x).$$

Thus,

$$H(x) = \exp(-2\ln(1-x)) = \frac{1}{(1-x)^2} = \left(\frac{1}{1-x}\right)' = \left(\sum_{n=0}^{\infty} x^n\right)' = \sum_{n=0}^{\infty} (n+1)x^n = \sum_{n=0}^{\infty} (n+1)! \cdot \frac{x^n}{n!}.$$

Thus, $h(n) = (n+1)!$.

Suppose that $f(n)$ is the number of ways to carry out some task on $[n]$ and let $g(n)$ be the number of ways to carry out some other task on $[n]$. When defining $h(n)$, using the exponential generating function means that the elements of $[n]$ are distinguishable, while using the ordinary generating function means that the elements of $[n]$ are indistinguishable. In other words, the exponential generating function uses subsets of $[n]$, while the ordinary generating function uses intervals. We can use this to get the following analogs.

Now, if $h(n) = \sum_{i=0}^n f(i)g(n-i)$, then the ordinary generating functions satisfy $H(x) = F(x) \cdot G(x)$, which is the analog of the product formula. Similarly, if

$$h(n) = \sum_{\lambda \vdash n} f(\lambda_1) \cdots f(\lambda_k) g(k),$$

then the ordinary generating functions satisfy $H(x) = G(F(x))$, which is the analog of the compositional formula. Assuming that $g(n) = 1$, we get the analog of the exponential formula to get that $H(x) = \frac{1}{1-F(x)}$.

Definition 16.2. A permutation w is down-up if it satisfies $w(1) > w(2) < w(3) > w(4) < \cdots$. A permutation w is up-down if it satisfies $w(1) < w(2) > w(3) < w(4) > \cdots$.

Lemma 16.3. *There exists a bijection between down-up and up-down permutations.*

Proof. Send w to $\bar{w}$, given by $\bar{w}(j) = n+1-j$. ■

We let E_n denote the number of down-up permutations of $[n]$. Let us consider $F(x) = \sum_{n=0}^{\infty} E_n \cdot \frac{x^n}{n!}$. Let w be a down-up or an up-down permutation of $[n+1]$. Let $l+1$ be the position of $n+1$ in w , and let S be the set of numbers to the left of $n+1$ in this permutation, and $\bar{S}$ be the set of set of numbers to the right of $n+1$. Let u be the permutation of S and v be the permutation of $\bar{S}$. Then v and the reverse of u are up-down permutations of S . There are $\binom{n}{k}$ choices for S , E_k choices for u , and E_{n-k} choices for v . Thus,

$$2E_{n+1} = \sum_{k=0}^n \binom{n}{k} E_k E_{n-k}.$$

Now, $F(x) = \sum_{n=0}^{\infty} E_n \frac{x^n}{n!}$. We have that

$$2F'(x) = \sum_{n=0}^{\infty} 2E_{n+1} \frac{x^n}{n!} = F(x)^2 + 1,$$

so

$$2F'(x) = F(x)^2 + 1$$

with $F(0) = 1$. The unique solution is given by $\sec(x) + \tan(x)$.

17 Lecture 17

Recall that S_n is the set of permutations of $[n]$.

Definition 17.1. For $w \in S_n$, we say that $i \in [n-1]$ is a descent of w if $w_i > w_{i+1}$. We write $D(w)$ for the set of descents of w , and $\text{des}(w) = |D(w)|$.

We write $A(n, k) = |\{w \in S_n : \text{des}(w) = k-1\}|$.

Proposition 17.2. $A(n, k) = k \cdot A(n-1, k) + (n-k+1)A(n-1, k-1)$.

Proof. Let $w \in S_n$, and remove n from the list given by writing out $w(1), \dots, w(n)$. Either $\text{des}(w)$ stays the same or it drops by 1. If it stays the same, either n is at the end, or there are $x > y$ with $w(i-1) = x$, $w(i) = n$ and $w(i+1) = y$. In the latter case, we get a permutation of $[n-1]$ with $k-1$ descents, so we have $(k-1)A(n-1, k)$ choices. In the former case, there are $A(n-1, k)$ choices. Thus, this case contributes $kA(n-1, k)$ choices. Now suppose that the number of descents drops by 1. Either n is first, or there are $x < y$ with $w(i-1) = x$, $w(i) = n$ and $w(i+1) = y$. In the second case, there are $n-k$ choices for inserting n in $A(n-1, k-1)$ permutations, and in the first case, there are $A(n-1, k-1)$ permutations. Thus, this contributes $(n-k+1)A(n-1, k-1)$. ■

Write $A_n(x) = \sum_{k=1}^n x^k A(n, k)$ for $n \geq 1$, and $A_0(x) = 1$. Then, we have that $A_n(x) = \sum_{w \in S_n} x^{1+\text{des}(w)}$.

Proposition 17.3. For each $n \geq 0$,

$$\sum_{m=0}^{\infty} m^n x^m = \frac{A_n(x)}{(1-x)^{n+1}}.$$

Proof. We prove this by induction on n . When $n = 0$, this is clearly true. Now suppose that this holds for some $n \geq 0$. Take the derivative and multiply by x . The left hand side becomes

$$\sum_{m=1}^{\infty} m^{n+1} x^m = \sum_{m=0}^{\infty} m^{n+1} x^m.$$

The right hand side becomes

$$x \cdot \frac{A'_n(x)(1-x)^{n+1} + A_n(x)(n+1)(1-x)^n}{(1-x)^{2n+2}}.$$

Thus, we have that

$$\sum_{m=0}^{\infty} m^{n+1} x^m = \frac{x(1-x)A'_n(x) + (n+1)xA_n(x)}{(1-x)^{n+2}}.$$

It remains to show that $A_{n+1}(x) = x(1-x)A'_n(x) + (n+1)xA_n(x)$. Taking the coefficient of x^k on the left, we get $kA(n, k) + (n-k+2)A(n, k-1) = A(n+1, k)$. ■

Theorem 17.4.

$$\sum_{n=0}^{\infty} A_n(x) \cdot \frac{t^n}{n!} = \frac{1-x}{1-x \cdot e^{(1-x)t}}.$$

Proof. Notice that

$$\sum_{n=0}^{\infty} \frac{A_n(x)}{(1-x)^{n+1}} \frac{y^n}{n!} = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} m^n x^m \frac{y^n}{n!} = \sum_{m=0}^{\infty} x^m \sum_{n=0}^{\infty} \frac{(my)^n}{n!} = \sum_{m=0}^{\infty} x^m e^{my} = \frac{1}{1-xe^y}.$$

Substitute $y = t(1-x)$ to get

$$\sum_{n=0}^{\infty} \frac{A_n(x)}{1-x} \frac{t^n}{n!} = \frac{1}{1-xe^{(1-x)t}}.$$

Thus,

$$\sum_{n=0}^{\infty} A_n(x) \cdot \frac{t^n}{n!} = \frac{1-x}{1-xe^{(1-x)t}}.$$

■

18 Lecture 18

Definition 18.1. Given $w \in S_n$, we define

$$\text{inv}(w) = |\{1 \leq i < j \leq n : w_i > w_j\}|.$$

For $i < j$, if $w_i > w_j$, we say (w_i, w_j) is an inversion of w .

Proposition 18.2. For $w \in S_n$, $\text{inv}(w) = \text{inv}(w^{-1})$.

Proof. (w_i, w_j) is an inversion of w if and only if (i, j) is an inversion of w^{-1} . ■

We wish to compute

$$\sum_{w \in S_n} q^{\text{inv}(w)}.$$

Definition 18.3. The q -analog of n is

$$[n]_q = 1 + q + q^2 + \cdots + q^{n-1}.$$

The q -factorial of n is

$$[n]_q! = [1]_q \cdot [2]_q \cdots [n]_q.$$

These go to n and $n!$ as q goes to 1 respectively.

Now, as q goes to 1, $\sum_{w \in S_n} q^{\text{inv}(w)}$ also tends to $n!$.

Theorem 18.4.

$$\sum_{w \in S_n} q^{\text{inv}(w)} = [n]_q!.$$

Proof. Let $w \in S_n$. Let a_i be the number of j 's to the left of i such that $j > i$. Then, $\text{inv}(w) = a_1 + \cdots + a_n$. To see this, notice that w_i appears in the right of an inversion a_i times. Now, for each i , $0 \leq a_i \leq n - i$. We further claim that w is uniquely determined by $(a_1, a_2, \dots, a_n)$. Start with $u^{(n+1)}$, which we define to be the empty sequence. For $i = n, n-1, \dots, 1$, let $u^{(i)}$ to be obtained by inserting i into $u^{(i+1)}$ so that it has a_i terms to the left. This gives a bijection between S_n and the set of sequences $(a_1, \dots, a_n)$ satisfying $0 \leq a_i \leq n - i$. Now,

$$\begin{aligned} \sum_{w \in S_n} q^{\text{inv}(w)} &= \sum_{\substack{(a_1, \dots, a_n) \\ 0 \leq a_i \leq n-i}} q^{a_1 + \cdots + a_n} = \sum_{a_1=0}^{n-1} \sum_{a_2=0}^{n-2} \cdots \sum_{a_n=0}^0 q^{a_1 + \cdots + a_n} \\ &= \sum_{a_1=0}^{n-1} q^{a_1} \sum_{a_2=0}^{n-2} q^{a_2} \cdots \sum_{a_n=0}^0 q^{a_n} \\ &= ([n]_q)([n-1]_q) \cdots ([1]_q) = [n]_q!. \end{aligned}$$

■

Definition 18.5. The major index of $w \in S_n$ is

$$\text{maj}(w) = \sum_{i \in D(w)} i.$$

Theorem 18.6.

$$\sum_{w \in S_n} q^{\text{maj}(w)} = [n]_q!.$$

Proof. Recall that S_n is in bijection with sequences $(a_1, \dots, a_n)$ satisfying $0 \leq a_i \leq n - i$. We want to find $w \in S_n$ with $\text{maj}(w) = a_1 + \dots + a_n$. By previous computations, this proves the result. Given a sequence $u = u_1 u_2 \dots u_k$, define

$$\text{maj}(u) = \sum_{\substack{i \in [k-1] \\ u_i > u_{i+1}}} i.$$

Let $u^{(n+1)}$ be the empty sequence. For $i = n, n-1, \dots, 1$, let $u^{(i)}$ be obtained from $u^{(i+1)}$ by inserting i so that $\text{maj}(u^{(i)}) - \text{maj}(u^{(i+1)}) = a_i$. Such an insertion place exists and is unique provided that $0 \leq a_i \leq n - i$. We show this for $w = w_1 \dots w_n$ inserting 0. Let the descent set of w be $D(w) = \{d_1 < d_2 < \dots < d_k\}$. If we insert 0 at d_k , then the major index does not change. If we insert it at d_{k-1} , the major index increases by 1. Similarly, inserting 0 at d_{k-i} , the major index increases by i . Now, write $[n] \setminus D(w) = \{a_1 < a_2 < \dots < a_{n-k} = n\}$. If we insert 0 at $a_{n-k} = n$, the major index increases by n . If we insert it at a_{n-k-1} , then there are $n - a_{n-k-1} - 1$ descents to the right of 0, so the major index increases by $a_{n-k-1} + n - a_{n-k-1} - 1 = n - 1$. Similarly, if we insert 0 at a_{n-k-j} , then there are $n - a_{n-k-j} - j$ descents to the right of 0, so the major index changes by $a_{n-k-j} + n - a_{n-k-j} - j = n - j$. Thus, each value is achieved exactly once. $\blacksquare$

The above theorem can be restated to say that there is a bijection $\phi : S_n \rightarrow S_n$ satisfying $\text{inv}(\phi(w)) = \text{maj}(w)$ for all $w \in S_n$.

19 Lecture 19

We say that $w \in S_n$ is even if $\text{inv}(w)$ is even, and that $w \in S_n$ is odd if $\text{inv}(w)$ is odd.

Lemma 19.1. *The number of odd permutations is equal to the number of even permutations.*

Proof. Swapping 1 and 2 gives a bijection between odd and even permutations. ■

Definition 19.2. Given $w \in S_n$, we define the permutation matrix A_w to be given by

$$a_{i,j} = \begin{cases} 1 & w_i = j \\ 0 & w_i \neq j \end{cases}.$$

By the definition of determinant, $\det(A_w) = (-1)^{\text{inv}(w)}$.

For $v, w \in S_n$, we define $v \cdot w = w \circ v$, so that $(vw)(i) = w(v(i))$.

Lemma 19.3. *If $v, w \in S_n$, then $A_{vw} = A_v A_w$.*

Proof. We have that

$$(A_v A_w)_{i,k} = \sum_{j=1}^n (A_v)_{i,j} (A_w)_{j,k},$$

which is nonzero (and equal to 1) if and only if $v(i) = j$ and $w(j) = k$. Then, $(vw)(i) = k$. ■

As $\det(AB) = \det(A) \det(B)$, if we write $\varepsilon_w = (-1)^{\text{inv}(w)}$, then $\varepsilon_v \cdot \varepsilon_w = \varepsilon_{vw}$. In other words, the product of two odd permutations and the product of two even permutations are even, while the product of an odd permutation with an even permutation or an even permutation with an odd permutation is odd.

Definition 19.4. Let $w \in S_n$. Choose $i \in [n]$, and find the smallest $r \geq 1$ such that $w^r(i) = i$. Then $(i, w(i), w^2(i), \dots, w^{r-1}(i))$ is called a cycle of w .

Now, the cycles are disjoint, have union $[n]$, and is defined up to cyclic rotation. We write w as a decomposition of cycles. For example, $w = (13)(2)(465)$ means that 1, 3, 2, and 4, 6, 5 are the cycles of w . The usual convention is to require that each cycle starts with the largest element. We omit terms of the form (a) , and it is understood that a is fixed by w .

Definition 19.5. A transposition is a permutation of the form (ij) for $i \neq j$.

20 Lecture 20

Lemma 20.1. *Any transposition is an odd permutation.*

Proof. Firstly, the case $(i \ i+1)$ is odd. Now, we have that $(ij) = (i \ i+1)(i+1 \ i+2) \cdots (j-2 \ j-1)(j-1 \ j)(j-1 \ j-2) \cdots (i+2 \ i+1)(i+1 \ i)$. This writes (ij) as a product of $2(j-i)-1$ transpositions of the form $(a \ a+1)$, so (ij) is an odd permutation. ■

Proposition 20.2. *A cycle $(a_1 a_2 \dots a_k)$ is an even permutation if and only if k is odd.*

Proof. We prove this by induction on k , with $k = 2$ being established. Now, we have that $(a_1 a_2 \dots a_k) = (a_{k-1} a_k)(a_1 a_2 \dots a_{k-1})$. ■

Definition 20.3. The Signless Stirling number of the first kind is given by $c(n, k) = |\{w \in S_n : w \text{ has } k \text{ cycles}\}|$.

For example, $c(n, n)$ is the number of permutations with n cycles, which is 1, and $c(n, 1)$ is the number of permutations with 1 cycle, which is $(n-1)!$.

Proposition 20.4. *For $1 \leq k \leq n$, $c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k)$.*

Proof. Let $w \in S_n$. Look at the cycle containing n , and remove n from it. If this cycle is (n) , there are $c(n-1, k-1)$ options. If this cycle has length greater than 1, then there are $c(n-1, k)$ options, but we have $n-1$ places into which we can insert n . ■

Theorem 20.5.

$$\sum_{k=1}^n c(n, k) x^k = x(x+1)(x+2) \cdots (x+n-1).$$

Proof. We prove this by induction on n . When $n = 1$, both sides are x . Now suppose that this holds for $n-1$,

$$\sum_{k=1}^{n-1} c(n-1, k) x^k = (x+n-2)_{n-1}.$$

Multiply both sides by $x+n-1$. Then the left hand side is

$$\sum_{k=1}^n ((n-1)c(n-1, k) + c(n-1, k-1)) x^k = \sum_{k=1}^n c(n, k) x^k.$$

■

Recall that we have

$$\sum_{k=0}^n S(n, k)(x)_k = x^n.$$

Definition 20.6. The Stirling Numbers of the first kind are given by $s(n, k) = (-1)^{n-k} c(n, k)$.

Corollary 20.7.

$$\sum_{k=1}^n s(n, k) x^k = (x)_n.$$

Proof. Multiply our formula for $c(n, k)$ by $(-1)^n$ and substitute $-x$ for x . ■

Thus, we have the dual equations

$$\sum_{k=1}^n s(n, k)x^k = (x)_n$$

$$\sum_{k=0}^n S(n, k)(x)_k = x^n$$

Thus, if we consider the two bases of polynomials of degree at most N given by $1, x, x^2, \dots, x^N$ and $1, (x)_1, (x)_2, \dots, (x)_N$, then these two matrices are inverses of each other.

21 Lecture 21

Definition 21.1. For $w \in S_n$, we let $c_i(w)$ be the number of cycles of w of length i , and we write $\text{type}(w) = (c_1(w), c_2(w), \dots, c_n(w))$.

Notice that we have

$$n = \sum_{i=1}^n i c_i(w).$$

Proposition 21.2. Suppose that $c = (c_1, c_2, \dots, c_n)$ satisfies

$$n = \sum_{i=1}^n i c_i.$$

Then, the number of $w \in S_n$ with $\text{type}(w) = c$ is $\frac{n!}{1^{c_1} \cdot c_1! \cdot 2^{c_2} \cdot c_2! \cdot \dots \cdot n^{c_n} \cdot c_n!}$.

Proof. Let $w = w_1 w_2 \dots w_n \in S_n$. Insert parenthesis into w to obtain $w' \in S_n$ of $\text{type}(w') = c$ so that cycle lengths are weakly increasing. To find the number of such w' , for each i , we divide by $c_i!$, as we can permute the i -cycles these many ways, and divide by i^{c_i} , as each cycle has i equivalent permutations. Thus, the total number of such permutations is $\frac{n!}{1^{c_1} \cdot c_1! \cdot 2^{c_2} \cdot c_2! \cdot \dots \cdot n^{c_n} \cdot c_n!}$. ■

The canonical cycle notation is that we write the maximal element of each cycle first, and the cycles are ordered so that the maximal elements increase from left to right.

Lemma 21.3 (Foata Transition Lemma). Let $w \in S_n$ be written in canonical cycle notation, and let $f(w) \in S_n$ be obtained by removing all parentheses. Then f is a bijection.

Definition 21.4. Let $v \in S_n$, and write $v = v_1 v_2 \dots v_n$. Then v_j is a left-to-right maximum if $v_i < v_j$ for all $1 \leq i < j$.

22 Lecture 22

Corollary 22.1. *The number of $v \in S_n$ with k left-to-right maxima is $c(n, k)$.*

Proposition 22.2. *Fix $i \neq j$, and choose a random $w \in S_n$. Then the probability that i and j are in the same cycle of w is $1/2$.*

Proof. Notice that the answer does not depend on our choice of i and j . Thus, choose $i = n - 1$ and $j = n$. Choose a uniformly random $v \in S_n$, and let w be the inverse of v in the Foata Transition Lemma. Then w is also a uniformly random element of S_n . Now, if $n - 1$ comes before n in v , then $n - 1$ and n are not in the same cycle in w . If $n - 1$ comes after n in v , then they are in the same cycle in w . Because v was chosen at random, the probability of this happening is $1/2$. ■

Theorem 22.3 (Permutation Exponential Formula). *Let $f(n)$ and $h(n)$ be sequences with $f(0) = h(0) = 0$. Suppose that*

$$h(n) = \sum_{w \in S_n} f(|C_1|) \cdots f(|C_k|),$$

where the C_i are cycles of w . Then, we have

$$H(x) = \exp \left(\sum_{n=1}^{\infty} f(n) \frac{x^n}{n} \right).$$

Proof. Let $g(n) = f(n) \cdot (n - 1)!$. Then,

$$h(n) = \sum_{w \in S_n} f(|C_1|) \cdots f(|C_k|) = \sum_{\{S_1, \dots, S_k\} \in \Pi([n])} g(|S_1|) \cdots g(|S_k|),$$

as $g(|S_i|)$ is the number of cyclic orders on S_i times $f(|S_i|)$. Using the usual exponential formula, we get the theorem. ■

Definition 22.4. $w \in S_n$ is an involution if w^2 is the identity.

In other words, involutions are the permutations where all cycles have length 1 or 2. Thus, let

$$f(n) = \begin{cases} 0 & n = 0 \\ 1 & n = 1, 2 \\ 0 & \text{else} \end{cases}.$$

Then,

$$h(n) = \sum_{w \in S_n} f(n)$$

is the number of involutions. Now,

$$H(x) = \exp \left(f(1) \cdot \frac{x^1}{1} + f(2) \cdot \frac{x^2}{2} \right) = \exp \left(x + \frac{x^2}{2} \right) = \exp(x) \exp \left(\frac{x^2}{2} \right).$$

Expanding, this is

$$\left(\sum_{k=0}^{\infty} \frac{x^k}{k!}\right) \left(\sum_{i=0}^{\infty} \frac{x^{2i}}{2^i i!}\right).$$

Now, the coefficient of $x^n/n!$ is

$$\sum_{i=0}^{\lfloor n/2 \rfloor} \frac{1}{2^i i!} \frac{1}{(n-2i)!} \cdot n! = \sum_{i=0}^{\lfloor n/2 \rfloor} \frac{(2i)!}{2^i i!} \binom{n}{2i}.$$

We write $(2i-1)!! = (2i)!/2^i i!$, so that

$$h(n) = \sum_{i=0}^{\lfloor n/2 \rfloor} (2i-1)!! \binom{n}{2i},$$

where we set $(-1)!! = 1$.

23 Lecture 23

Theorem 23.1. *Let*

$$H(x, t) = \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} c(n, k) t^k \cdot \frac{x^n}{n!}.$$

Then,

$$H(x, t) = (1 - x)^{-t}.$$

Proof. Let $f(n) = t$ for all $n \geq 1$, and define

$$h_t(n) = \sum_{w \in S_n} f(|C_1|) \cdots f(|C_k|) = \sum_{k=0}^{\infty} c(n, k) t^k,$$

where the C_i are the cycles of w . Then $H(x, t)$ is the exponential generating function for $h_t(n)$, so

$$H(x, t) = \exp \left(\sum_{n=1}^{\infty} f(n) \frac{x^n}{n} \right) = \exp(-t \ln(1 - x)) = (1 - x)^{-t}.$$

■

Definition 23.2. Let $\pi \in S_k$. A permutation $w \in S_n$ is π -avoiding if there is no $1 \leq i_1 < \cdots < i_k \leq n$ such that $w_{i_{\pi(1)}} < w_{i_{\pi(2)}} < \cdots < w_{i_{\pi(k)}}$.

For any $\pi \in S_3$, the number of π -avoiding permutations in S_n is C_n .

Notation. Let V be a set. We write $\binom{V}{k}$ for the set of k -element subsets of V .

A graph is a pair $G = (V, E)$, where V is the set of vertices, and $E \subseteq \binom{V}{2}$ is the set of edges. We will assume that G is simple, so that G has no multiple edges, and no edges connecting a vertex to itself. A walk in G is a sequence $e_1 e_2 \dots e_k$ of edges such that each e_i and e_{i+1} share an endpoint. A path in G is a walk which goes through any vertex at most once. A cycle is a path with the same starting and ending points, where no other vertices are repeated. We say G is connected if any two vertices can be connected by a walk.

For example, the complete graph K_n is the graph with $V = [n]$ and $E = \binom{[n]}{2}$. The complete bipartite graph $K_{m,n}$ is the graph with $V = \{u_1, \dots, u_m, v_1, \dots, v_n\}$ and $E = \{\{u_i, v_j\} : 1 \leq i \leq m, 1 \leq j \leq n\}$. The path graph has $V = [n]$ and $E = \{\{1, 2\}, \{2, 3\}, \dots, \{n-1, n\}, \{n, 1\}\}$.

A tree is a connected graph without cycles.

Proposition 23.3. *Let $G = (V, E)$ be a connected graph. Then the following are equivalent.*

- (1) G is a tree.
- (2) For any $x, y \in V$, there is a unique path from x to y .
- (3) If you remove any edge from G , it is no longer connected.
- (4) $|V| = |E| + 1$.

24 Lecture 24

Given graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, we say $G_1 = G_2$ if and only if $V_1 = V_2$ and $E_1 = E_2$.

Let us consider how many different trees there are with vertex set $[n]$. When $n = 2$, there is 1 tree, with $E = \{\{1, 2\}\}$. When $n = 3$, there are 3 trees, with

$$E = \{\{1, 2\}, \{2, 3\}\}, \{\{1, 2\}, \{1, 3\}\}, \{\{1, 3\}, \{2, 3\}\}.$$

When $n = 4$, there are 16 trees (there are $4!/2$ ways to choose the edges if the graph is a “line”. and 4 ways if the graph has a central vertex). When $n = 5$, there are 125 trees (along with the previous two patterns, there is a graph in the shape of a ‘Y’, which adds another $5!/2$ trees).

Notation. We write $\mathbb{T}(n)$ for the set of trees on $[n]$.

Theorem 24.1 (Cayley’s Formula). *The number of different trees on $[n]$ is $|\mathbb{T}(n)| = n^{n-2}$.*

Proof. We show that $n^2 \cdot |\mathbb{T}(n)| = n^n$. The left hand side counts triples (T, ℓ, r) , where $T \in \mathbb{T}(n)$, and $\ell, r \in [n]$, and the right hand side counts functions from $[n]$ to itself. Take any such function $f : [n] \rightarrow [n]$. Draw it as a directed graph, where $i \in [n]$ points towards $f(i) \in [n]$. Here, each vertex has exactly one outgoing arrow. Now, if we apply f enough times, we will have $f^k(i) = f^\ell(i)$ for some $k \neq \ell$. This allows us to find a “cycle” in the graph. Let $C \subseteq [n]$ be the set of vertices which belong to such a “cycle”, so that $f|_C : C \rightarrow C$ is a bijection. Write $C = \{c_1 < c_2 < \dots < c_k\}$, and write $f|_C$ in two-line notation. Let $\ell = f(c_1)$ and $r = f(c_k)$. Define T as follows: draw a path from $f(c_1)$ to $f(c_2)$ to $f(c_3)$ to $\dots$ to $f(c_k)$. For each $i \notin C$, draw an edge connecting i to $f(i)$. We claim that T is a tree. T has $n - 1$ edges, and $n = (n - 1) + 1$ vertices, so we only need to show that T is connected. The path $f(c_1), \dots, f(c_k)$ is connected, and for each $i \notin C$, the sequence $i, f(i), f(f(i)), \dots$ eventually terminates at the path from ℓ to r . We claim that this map is a bijection. Given (T, ℓ, r) , we wish to recover f . There is exactly one path in T from ℓ to r , which recovers C and the one-line notation of $f|_C$. For each $i \notin C$, T contains a unique shortest path from i to C . Let $f(i)$ be the next vertex after i on this path. Clearly the maps are inverses of each other. ■

Definition 24.2. Given $G = (V, E)$ and $v \in V$, the degree $d_v(G) = d_v$ is the number of neighbors of v in G . A leaf is a vertex of degree 1.

Lemma 24.3 (Handshake Lemma). *For $G = (V, E)$, $\sum_{v \in V} d_v = 2|E|$.*

Definition 24.4. Fix $n \geq 1$. We define

$$F_n(x_1, x_2, \dots, x_n) = \sum_{T \in \mathbb{T}(n)} x_1^{d_1(T)} x_2^{d_2(T)} \dots x_n^{d_n(T)}.$$

For example, $F_1(x_1) = 1$, $F_2(x_1, x_2) = x_1 x_2$, $F_3(x_1, x_2, x_3) = x_1 x_2 x_3 (x_1 + x_2 + x_3)$, and $F_4(x_1, x_2, x_3, x_4) = x_1 x_2 x_3 x_4 (x_1 + x_2 + x_3 + x_4)^2$.

25 Lecture 25

Theorem 25.1.

$$F_n(x_1, x_2, \dots, x_n) = x_1 x_2 \cdots x_n (x_1 + x_2 + \cdots + x_n)^{n-2}.$$

In particular, the number of trees with degree sequence $d_1, d_2, \dots, d_n$ is

$$\binom{n-2}{d_1-1, d_2-1, \dots, d_n-1},$$

which is nonzero by the handshake lemma.

Proof. We use Prüfer sequences. A Prüfer sequence is a sequence $a = (a_1, \dots, a_{n-2})$ of elements of $[n]$. We aim to find a bijection sending a tree T to $a(T) = (a_1, \dots, a_{n-2})$ so that, for each $i \in [n]$, i appears in $a(T)$ exactly $d_i(T) - 1$ times. The theorem follows by rewriting the sum in terms of Prüfer sequences. Given a tree T , find the smallest labeled leaf $i \in [n]$. Let $j \in [n]$ be the unique neighbor of i . Append j to $a(T)$ and remove i from T . Iterate this process until T has 2 vertices. Then each i appears $d_i(T) - 1$ times. Now suppose that we are given a sequence $a = (a_1, a_2, \dots, a_{n-2})$. Reconstruct the degrees $(d_1, d_2, \dots, d_n)$. We thus know which vertices are leaves, and what the minimal leaf $i \in [n]$ is. We can then connect i to a_1 , and reconstruct the rest of the tree recursively. ■

Definition 25.2. A spanning tree of $G = (V, E)$ is a tree $T = (V, E')$ for some $E' \subseteq E$.

For example, $\mathbb{T}(n)$ is the set of spanning trees of K_n .

Theorem 25.3. The number of spanning trees of $K_{m,n}$ is $m^{n-1}n^{m-1}$.

Definition 25.4. A forest is a graph without cycles. In other words, a forest is a disconnected union of trees. A rooted forest is a forest and a choice of a “root vertex” in each connected component.

Proposition 25.5. The number of rooted forests on $[n]$ is $(n+1)^{n-1}$.

Proof. Take a rooted forest on $[n]$. Add vertex $n+1$ and connect this to all roots to get a tree on $[n+1]$. This is a bijection. ■

26 Lecture 26

Consider a one-way street with n parking spots $1, 2, \dots, n$. Suppose that n cars enter the street. Fix $f : [n] \rightarrow [n]$, and suppose that car i prefers spot f_i . Each car proceeds to its preferred spot, and if it is occupied, takes the next available spot. We say that $f : [n] \rightarrow [n]$ is a parking function if all cars are able to park.

For example, all permutations are parking functions. So is the map $f(i) = 1$. However, $f(i) = 2$ is not a parking function.

Proposition 26.1. *Let $f : [n] \rightarrow [n]$. Let $a_1 \leq a_2 \leq \dots \leq a_n$ be the increasing rearrangement of the values of f . Then f is a parking function if and only if each $a_i \leq i$ for $i \in [n]$.*

Proof. Suppose that $a_i \leq i$ for all $i \in [n]$. Then $a_1 \leq 1$, so at least 1 car prefers spot 1. Now, $a_2 \leq 2$, so at least 2 cars prefer spots below 2, so the first two spots will be filled. Continuing like this, all spots will be filled. Conversely, suppose that f is a parking function. Then, for $i \in [n]$, if the i th car parks at g_i , then $f_i \leq g_i$. Thus, for each $j \in [n]$, $|\{i \in [n] : f_i \leq j\}| \geq j$, meaning that $a_j \leq j$. ■

Corollary 26.2. *Any permutation of the entries of a parking function is a parking function.*

Theorem 26.3. *The number of parking functions $f : [n] \rightarrow [n]$ is $(n+1)^{n-1}$.*

Proof. Consider $n+1$ parking spots arranged in a circle, and consider a function $f : [n] \rightarrow [n+1]$. Each car goes to spot $f(i)$, and then looks for the next available spot. All cars are now able to park, and exactly one spot j will remain empty. If all values of f are increased by $k \pmod{n+1}$, then the new empty spot is $j+k \pmod{n+1}$. Now, f is a parking function if and only if $j = n+1$. The total number of functions from $[n]$ to $[n+1]$ is $(n+1)^n$. Dividing by $n+1$, the number of parking functions is $(n+1)^{n-1}$. ■

Corollary 26.4. *If you randomly choose a function $f : [n] \rightarrow [n]$, the probability that it is a parking function is $\frac{(n+1)^{n-1}}{n^n} = \frac{1}{n} \left(1 + \frac{1}{n}\right)^{n-1}$. As n gets large, this tends to $\frac{e}{n}$.*

27 Lecture 27

Notation. Given a graph G , we write $\kappa(G)$ for the number of spanning trees of G .

Definition 27.1. Let $G = (V, E)$, with $V = [n]$. The Laplacian matrix $L(G)$ is given by

$$(L(G))_{i,j} = \begin{cases} -1 & \{v_i, v_j\} \in E \\ d_i & i = j \\ 0 & \text{else} \end{cases}.$$

Proposition 27.2. $\det L(G) = 0$.

Proof. The sum of the columns is zero. ■

Theorem 27.3 (Matrix-Tree Theorem). *Let $G = (V, E)$, and let $L_0(G)$ be obtained from $L(G)$ by removing the last row and last column. Then $\kappa(G) = \det(L_0(G))$.*

Definition 27.4. A rooted plane tree is a tree drawn in the plane with a chosen root vertex R connected to $(v_1, v_2, \dots, v_k)$ (where order matters), where each v_i is a root of a smaller rooted plane tree. In other words, a rooted plane tree is an ordered tuple of smaller rooted plane trees $T = (T_1, T_2, \dots, T_k)$.

Definition 27.5. $T = (T_1, T_2, \dots, T_k)$ and $T' = (T'_1, T'_2, \dots, T'_m)$ are isomorphic as rooted plane trees if and only if $k = m$ and $T_i \cong T'_i$ as rooted plane trees.

Theorem 27.6. *The number of rooted plane trees on $n + 1$ vertices is C_n .*

Proof. We find a bijection that maps T to $P(T)$, a path from $(0, 0)$ to (n, n) staying above $y = x$. Let $T = (T_1, T_2, \dots, T_k)$. We define $P(T)$ recursively. If $k > 1$, $P(T)$ touches the diagonal $k - 1$ times. Let T'_i be obtained from T_i by adding an extra root. Then each T'_i has strictly less vertices than T . Set $P(T)$ to be the concatenation of $P(T'_1), \dots, P(T'_k)$. If $k = 1$, $P(T)$ is given by a north step, followed by $P(T')$, followed by an east step. We show that this is a bijection. Suppose that we are given a path P . If P never touches the diagonal, then $P = (N, P', E)$ for some other path P' . Set T to be T' with an extra root. Otherwise, P touches the diagonal $k - 1$ times for some $k > 1$. Then P is a concatenation of k paths $(P_1, P_2, \dots, P_k)$, so set $T = (T_1, T_2, \dots, T_k)$, where T_i is computed from P_i . ■

28 Lecture 28

We convert $w \in S_n$ into a labeled rooted tree $T(w)$ as follows. Write w as LnR in one-line notation. Then $T(w)$ has a root labeled n which is connected to $T(L)$ and $T(R)$, which are subtrees computed recursively from L and R .

Definition 28.1. A Decreasing Binary Tree is a rooted tree T , such that each vertex is connected to a (possibly empty) left subtree and a (possibly empty) right subtree, and such that the label of each vertex is strictly greater than the labels of its children.

We write $\text{BT}(n)$ for the set of decreasing binary trees on $[n]$. We then have that $|\text{BT}(n)| = n!$, as the map that we described is a bijection.

Definition 28.2. A plane binary tree is a rooted tree (with unlabeled vertices) such that each vertex is connected to a (possibly empty) left subtree and a (possibly empty) right subtree. In other words, it is a pair $T = (T_L, T_R)$ of smaller plane binary trees.

We say that two plane binary trees $T = (T_L, T_R)$ and $T' = (T'_L, T'_R)$ are the same if and only if $T_L \cong T'_L$ and $T_R \cong T'_R$.

Theorem 28.3. The number of plane binary trees on n vertices is C_n .

Proof. Write p_n for the number of plane binary trees on n vertices. We show that p_n and C_n satisfy the same recurrence, as they agree when $n = 1$. Let T be a plane binary tree on $n + 1$ vertices, say $T = (T_L, T_R)$. Let i be the number of vertices in T_L . Then, we have p_i choices for T_L and p_{n-i} choices for T_R , so $p_{n+1} = \sum_{i=0}^n p_i p_{n-i}$. ■

Definition 28.4. The Narayana number $N(n, k)$ is given by

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}.$$

Notice that $N(n, k) = N(n, n + 1 - k)$.

Proposition 28.5.

$$N(n, 1) + N(n, 2) + \cdots + N(n, n) = C_n.$$